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linux convention: what return codes to use for "forcefully killed" / "aborted" / "cancelled"

In Linux and Unix-based systems, exit statuses (or return codes) are used by processes to indicate to the parent process how they ended. These statuses can indicate normal completion, various error conditions, or that the process was terminated by a signal. Here are the conventions related to "forcefully killed," "aborted," or "cancelled" processes:

1. Forcefully Killed (by a signal):

- When a process is forcefully killed, such as with the `kill` command, it is terminated by a signal. The exit code for a process that ends due to a signal is not directly the signal number but is calculated using a formula. By convention, if a process is terminated by signal `n`, the exit status is $128 + n$. For example, if a process is killed with `SIGKILL` (signal 9), the exit status would conventionally be $128 + 9 = 137$. Similarly, if terminated with `SIGTERM` (signal 15), the exit status would be $128 + 15 = 143$.

2. Aborted:

- An abort situation usually occurs due to some critical error during execution, such as an assertion failure or an unhandled exception. In C and C++ programs, this is often explicitly triggered by calling the `abort()` function, which causes the program to terminate with the `SIGABRT` signal (signal 6). The exit status for an aborted process would conventionally be $128 + 6 = 134$.

3. Cancelled:

- There isn't a specific universal exit code for a "cancelled" state, as cancellation is application-specific. It might be that a user action or some condition within the application triggers a graceful shutdown procedure. In these cases, the application should choose an appropriate exit code that signifies cancellation to the parent process or supervising system. It is common for applications to use specific non-zero exit codes (other than those reserved for signals) to indicate cancellation or specific error conditions. The actual value might depend on the application's documentation or the developer's design.

It's essential to note that while the convention for signal-induced terminations is relatively standard (i.e., `128 + signal number`), the handling and interpretation of exit codes for situations like "aborted" or "cancelled" can vary between applications. Developers often document these codes in their application manuals or help pages, so users and other programs can understand what each code means.